

$\Omega \neq \emptyset \rightarrow$ Sample space

$\mathcal{F} \subseteq 2^\Omega \rightarrow$ Set of events

$\hookrightarrow$ - In most cases, especially when

Ω is finite, we take $\mathcal{F} = 2^\Omega$

- In some cases $\mathcal{F}$ is indeed a strict subset of 2^Ω . For the time being we will not worry much about the implications of $\mathcal{F}$ being a strict subset of 2^Ω .

Probability $P: \mathcal{F} \rightarrow [0, 1]$ is a function

satisfying the following criteria:-

(i) $P(\emptyset) = 0$

(ii) $P(\Omega) = 1$

(iii) Let $A_1, A_2, \dots \in \mathcal{F}$ be a sequence of mutually exclusive events, i.e, for all $i \neq j$, $A_i \cap A_j = \emptyset$.

Then $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$.

An important consequence of the definition of probability

(iii') let $n \in \mathbb{N}$, and let $A_1, \dots, A_n \in \mathcal{F}$ be mutually exclusive. Then $P(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n P(A_i)$.

— To see this, define $A_j = \emptyset$ for all $j \geq n+1$.

Then, the sequence of events $A_i, i \geq 1$, continue to remain mutually exclusive.

By condition (iii),

$$P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i) \quad \dots \quad (1)$$

Now, since for $i \geq n+1$, $A_i = \emptyset$, we

$$\text{have } \bigcup_{i=1}^{\infty} A_i = \bigcup_{i=1}^n A_i$$

Also, by condition (i) $P(A_i) = 0$, for all $i \geq n+1$.

Plugging these in (D), we get

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i).$$

More consequences of the definition of probability

I. let $A, B \in \mathcal{F}$ such that $A \subseteq B$.

Then $P(A) \leq P(B)$.

Proof:~ Observe that

$$B = A \cup (B \setminus A) \quad \text{as } B \supseteq A.$$

Also, A and $B \setminus A$ are mutually exclusive.

$$\text{Then, } P(B) = P(A \cup (B \setminus A))$$

$$= P(A) + P(B \setminus A) \quad [\text{Condition (ii')}]$$

$$\geq P(A) \quad [\because P(B \setminus A) \geq 0]$$

. $\square$

II. let $A \in \mathcal{F}$. Then

$$P(A) + P(A^c) = 1$$

Proof:~ Note that

$$\Omega = A \cup A^c.$$

Then $1 = P(\Omega)$ [condition (ii)]

$$= P(A \cup A^c)$$

$$= P(A) + P(A^c) \text{ [condition (iii')]}]$$

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III. Inclusion-Exclusion formula

Let $A, B \in \mathcal{F}$. Then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Proof:~ Observe that

$$A \cup B = (A \cap B) \cup (A \setminus B) \cup (B \setminus A)$$

$$A = (A \setminus B) \cup (A \cap B)$$

$$B = (B \setminus A) \cup (A \cap B)$$

Also, note that $(A \setminus B)$, $(B \setminus A)$, $(A \cap B)$ are mutually exclusive.

Then,

$$P(A) + P(B)$$

$$= P(A \setminus B) + P(A \cap B) + P(B) + P(B \setminus A)$$

[condition (ii')]

$$= P(A \cap B) + P(A \cup B) \quad \text{[condition (ii')]}$$

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Remark :- In general, for n events $A_1, A_2, \dots, A_n$, the inclusion-exclusion formula is given by

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n (-1)^{i-1} \sum_{\substack{\{j_1, j_2, \dots, j_i\} \\ \subseteq \{1, 2, \dots, n\}}} P(A_{j_1} \cap A_{j_2} \cap \dots \cap A_{j_i})$$

In other words,

- For $n=3$

$$\begin{aligned} P(A_1 \cup A_2 \cup A_3) &= P(A_1) + P(A_2) + P(A_3) - P(A_1 \cap A_2) \\ &\quad - P(A_1 \cap A_3) - P(A_2 \cap A_3) + \\ &\quad P(A_1 \cap A_2 \cap A_3) \end{aligned}$$

- For $n=4$

$$\begin{aligned} P(A_1 \cup A_2 \cup A_3 \cup A_4) &= P(A_1) + P(A_2) + P(A_3) + P(A_4) \\ &\quad - P(A_1 \cap A_2) - P(A_1 \cap A_3) - P(A_1 \cap A_4) \\ &\quad - P(A_2 \cap A_3) - P(A_2 \cap A_4) - P(A_3 \cap A_4) \\ &\quad + P(A_1 \cap A_2 \cap A_3) + P(A_1 \cap A_2 \cap A_4) \\ &\quad + P(A_1 \cap A_3 \cap A_4) + P(A_2 \cap A_3 \cap A_4) \\ &\quad - P(A_1 \cap A_2 \cap A_3 \cap A_4) \end{aligned}$$

You can prove the inclusion-exclusion formula for general n via induction.

IV. Union bound / Boole's inequality

Let $A_1, \dots, A_n \in \mathcal{F}$. Then

$$P\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n P(A_i).$$

Proof:- We prove this by induction.

Consider the base case of $n=2$.

By the inclusion-exclusion formula

$$\begin{aligned} P(A_1 \cup A_2) &= P(A_1) + P(A_2) - P(A_1 \cap A_2) \\ &\leq P(A_1) + P(A_2) \end{aligned}$$

$$[\because P(A_1 \cap A_2) \geq 0]$$

So, the statement holds for $n=2$.

Now, assume the induction hypothesis that the statement holds for n , i.e.,

$$P\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n P(A_i).$$

Then $P\left(\bigcup_{i=1}^{n+1} A_i\right) = P\left(\bigcup_{i=1}^n A_i \cup A_{n+1}\right)$

$$\leq P\left(\bigcup_{i=1}^n A_i\right) + P(A_{n+1})$$

[Using the base case]

$$\leq \sum_{i=1}^n P(A_i) + P(A_{n+1})$$

[Using the induction hypothesis]

$$= \sum_{i=1}^{n+1} P(A_i).$$

Therefore, the statement holds for $n+1$, and

so, by the principle of mathematical

induction, the statement holds for all $n \geq 2$.

... $\square$

Remark:- Does the above prove that
for $A_1, A_2, \dots \in \mathcal{F}$

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} P(A_i) ?$$

No

The principle of mathematical induction
can only prove a statement for $n \in \mathbb{N}$,
and not for infinitely many 'objects'.

V. Let $A_1, A_2, \dots \in \mathcal{F}$. Then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} P(A_i).$$

Proof:- Define $B_1 = A_1$ and

$$B_i = A_i \setminus \left(\bigcup_{j=1}^{i-1} A_j\right)$$

Observe that $B_i \subseteq A_i$.

Claim:- $B_i \cap B_j = \emptyset$ if $i > j$.

To see this, $B_j \subseteq A_j \subseteq \bigcup_{l=1}^{i-1} A_l$

But $B_i = A_i \setminus \left(\bigcup_{l=1}^{i-1} A_l \right)$

So, $B_i \cap B_j = \emptyset$.

Thus, $B_1, B_2, \dots$ are mutually exclusive.

Claim :- $\bigcup_{i=1}^{\infty} B_i = \bigcup_{i=1}^{\infty} A_i$

To see this, let $x \in \bigcup_{i=1}^{\infty} B_i$.

Then, there exist j st. $x \in B_j$.

Then $x \in A_j$ and so $x \in \bigcup_{i=1}^{\infty} A_i$

So, $\bigcup_{i=1}^{\infty} B_i \subseteq \bigcup_{i=1}^{\infty} A_i$.

Let $y \in \bigcup_{i=1}^{\infty} A_i$.

Then, let j be the smallest index such that $y \in A_j$.

This means $y \notin \bigcup_{l=1}^{j-1} A_l$ as j is the smallest index satisfying $y \in A_j$.

Hence $y \in A_j \setminus \bigcup_{l=1}^{j-1} A_l = B_j$.

So $y \in \bigcup_{i=1}^{\infty} B_i$.

Hence $\bigcup_{i=1}^{\infty} A_i \subseteq \bigcup_{i=1}^{\infty} B_i$ and so,

$$\bigcup_{i=1}^{\infty} A_i = \bigcup_{i=1}^{\infty} B_i$$

Then,

$$\begin{aligned} P\left(\bigcup_{i=1}^{\infty} A_i\right) &= P\left(\bigcup_{i=1}^{\infty} B_i\right) \\ &= \sum_{i=1}^{\infty} P(B_i) \quad [\because B_i \text{'s are mutually exclusive and condition (ii)}] \\ &\leq \sum_{i=1}^{\infty} P(A_i) \quad [\because B_i \subseteq A_i, \text{ and hence } P(B_i) \leq P(A_i)] \end{aligned}$$

□